APPENDIX: PROJECT CANVAS

Title: Propaganda radar
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Workshop # : 3

MOTIVATION

customer: CRTA is an organization that want to promote transparency in state media influence

end-user: probably someone who either enters the site via link (from social media or similar), so probably critical of the state influence; or maybe anyone who has interest in the relationship between media, the government and politics

benefit to CRTA: more effective reporting (real-time, automated)

benefit to end-user: politologists or other researchers get more data, to support a bigger picture; casual visitor gets a more interactive and real-time experience

DATA COLLECTION

Selected government-controlled media sources: headlines, articles from their web sites. Via RSS, then scrape with site-specific custom scrapers

Data management plan:

Scraped data will live in the analysis backend, not exposed

from the output of the analysis, render dashboards (classification of news items) etc.

For test and training data, we'll ask CRTA for a hand-labelled data set

PREPROCESSING

Scraped data needs to be cleaned up for the analysis.

Goal is to classify the headlines (later also articles) into pre-specified buckets (classes like "personality cult building"). Initial buckets come from CRTA's article. Based on EDA we might need to add more buckets.

Apply common text cleaning and simplification steps (stemming etc.), learn to associate keywords and phrases to the classes.

EXPLORATORY DATA ANALYSIS (EDA)

For "live feed", or classifying a random article, it will be important to evaluate how good the coverage of the buckets is.

Figure out the rough proportion of each class by random sampling from the web site.

This has implications to training: if there's a lot of items that are outside the buckets, we need to collect more "random samples" ourselves, perhaps add more buckets

VISUALIZATIONS

Visualize the feature vector outputs by the classifier (vs hand-labelled ground truth), maybe some kind of PCA plot vs labels

Word clouds corresponding to each bucket

Clustering that would support or reject the current buckets (do we need more? Are some redundant or empty?), after preprocessing with BERT

LEARNING TASK (focus on problem definition)

Problem: Classifying pieces of text to pre-defined classes

Supervised learning: learn classes from hand-labelled data

Input: headline (later a full article)

Output: vector of probabilities of belonging to each class

Learning target: highest probability corresponds to the correct label

LEARNING APPROACH (focus on solution implementation)

Classification: Logistic Regression, Random Forest, XGBoost, SVM

Evaluation metric: F1 score

Hierarchical classification: is it any of the buckets? Vilifying vs glorifying? Consider training separate classifiers to each step.

For training data: hand-labeled data from CRTA + own scraping

COMMUNICATION OF RESULTS

Classifications of the past articles: lists of the classification results (buckets + the corresponding headlines)

Live feed (chronological list) of the news items after they've been analyzed: classification (later also the specific keywords or phrases that support the classification)

DATA PRIVACY AND ETHICAL CONSIDERATIONS (if applicable)

The classification buckets can be very negative: we should present the classification as a "weak claim" until it has been verified: either manually by a trusted moderator, or by the community. There must be a human in the loop.

All scraped raw text is public (no need for explicit consent), and we won't re-distribute it. Headlines can't be copyrighted.

There's no PII that wasn't included in the public text (no anonymization needed).

We don't collect data from the users.

ADDED VALUE

Politologists or other researchers get more data, to support a bigger picture

A casual visitor gets a more interactive and real-time experience

LEGEND

WEEK 1: Data collection/preprocessing

WEEK 2: EDA & visualizations

WEEKS 3-4: Machine/deep learning